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POLICY LEARNING AND INFORMATION PROCESSING DYNAMICS

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Policy Learning and Information Processing Dynamics

Abstract

Policy learning has long been heralded as one of the most versatile and widely recognized approaches to policy analysis, so much so that it is often viewed as an ontology of the policy process. However, its theoretical and analytical fineness notwithstanding, research offering a systematic approach to tracing the micro-foundations of policy learning remains rather embryonic. In this element, we begin with mapping the sprawling theoretical, conceptual, and analytical landscape of policy learning research. Leveraging this review, we develop a model of policy-embedded learning and information processing. Taking a Bayesian information processing perspective, this model enables analyses of how dynamics of attention, selection, and the interpretation of information and their interaction with existing belief structures influence learning outcomes. We then apply this model using public opinion survey data from the United States on the use of nuclear energy within the context of the Fukushima nuclear reactor incident. We conclude with potential avenues for future research, particularly focused on theoretically and analytically scaling up this model of policy-embedded learning and information process.

Keywords: Policy learning; Information Processing; Policy-embedded learning; Public Policy

1. Introduction to Policy Learning

1.1 Policy Learning: A Background

Why do policy actors do what they do? Why and how does a policy change or remain immutable? Those are some of the most quintessential puzzles in public policy research and practice. Several theoretical approaches and frameworks endeavored to solve these puzzles. For example, by focusing on changes in core beliefs (Sabatier & Jenkins-Smith, 1993), institutional arrangements (Ostrom, 1990), mechanisms of policy adjustment (Lindblom, 1959), patterns of policy change, and the role of external events (Baumgartner & Jones, 1993), or the role of policy entrepreneurs in coupling problems, solutions, and political conditions, (Kingdon, 1984), among others.

While these different theoretical approaches provided substantial insights into the particularities of policy change, the learning approach has maintained its position as perhaps one of the most fundamental and promising often heralded as a potent approach to explaining policy change and stability. The fundamental nature of policy learning has helped it proliferate across different theoretical approaches to the policy process. Accordingly, several of the theories and frameworks of the policy process discuss policy learning, at least implicitly. For example, within the Multiple Streams Approach, learning has been observed to occur in epistemic communities in the policy stream through changes in indicators in the problem stream and experimenting with various political strategies in the politics stream (Herweg et al., 2023). Furthermore, within the same framework, learning has also been linked to the key process by which policy entrepreneurs "couple" the different streams (Blum, 2018). Learning also crystallizes across different theories of the policy process, from policy diffusion and transfer (Volden et al., 2008) to the stages approaches to policymaking, from policy design to implementation (Biegelbauer, 2016; Zaki, 2024b). Additionally, policy learning has integrated into technical and social understandings of policy problems (Bennett & Howlett, 1992), policy advocacy (Zaki & Dupont, 2024), levels of policy change (Bennett & Howlett, 1992; Hall, 1993; Heclo, 1974; Moyson et al., 2017), and policy entrepreneurship (Blum, 2018). However, among the various policy process theories and frameworks, the Advocacy Coalition Framework (ACF), through its focus on the connections between belief systems, belief change, and policy-oriented learning, has one of the most substantive conceptualizations of the policy learning and policy change nexus (Bandelow et al., 2019; Jenkins-Smith & Sabatier, 1993; Nohrstedt et al., 2023; Pattison, 2018). Overall, policy learning has gained prominence across the board throughout policy analysis and policy process scholarship (Bennett & Howlett, 1992, 1992; Dunlop & Radaelli, 2022).

Apart from contributing to policy process theories, policy learning may constitute its own analytical framework (Dunlop & Radaelli, 2018a). Going further, policy learning may not only be a theoretical lens on the policy process, but it may also function as an ontology of policymaking. In other words, its nature is a fundamental process that shapes individual and collective behavior within the policy process (Kamkhaji, 2022). This stems from the background understanding of policy learning as a fundamental problem-solving process in which policy actors seek and process information and knowledge about policy problems, aiming to develop viable solutions while considering how this interacts with existing cognitive and institutional structures in which problem-solving occurs (Zaki et al., 2022). This process can lead to cognitive outcomes where policy actors update their knowledge and beliefs about policy problems or behavioral products, such as different degrees of policy change (Dunlop & Radaelli, 2013). Accordingly, an ontological view of policy learning ascribes that we can better explain policy behaviors and policy outcomes if we understand the key aspects of learning (i.e., who learns, what they learn, from whom, how, and why). In other words, we can better explain why policy actors behave in a certain way and the consequences of their actions as a function of their learning processes (Zaki, 2024c).

The theoretical status of policy learning and its role in shaping policymaking processes and outcomes are rooted in empirical accounts. For example, research shows that if policy actors learn optimally (i.e., from the right sources, at the right time, employing adequate modes of learning), they have higher chances of developing practical solutions to policy problems, delivering substantive reforms, or improving policy performance (Stark, 2019; Wai Yip So 2012). However, learning is not optimally approached when expertise, knowledge, and learning modes are misidentified or misused. In that case, policy learning can become misdirected, leading to policy failures or fiascos (Dunlop et al., 2020) or even value destruction (Zaki, 2024a). As a result, we increasingly see policy actors attempting to deliberately design and develop robust and effective policy learning processes (Zaki, 2024b). This manifests across a wide range of policy domains, from public health to agriculture and food security to crisis governance and beyond (Albright, 2011; Beach et al., 2021; Busetti & Righettini, 2023; Crow et al., 2023).

So, what makes policy learning so central to policy analysis? At the heart of the policy learning process lies the fuel of individual and collective behavior, that is, information and its processing, which constitute the raw material with which ideas, knowledge, beliefs, and social construction of policy problems are built (Nowlin, 2021; Zaki et al., 2022). Within learning, individuals and collectives exchange informational cues about policy problems and then reconcile and process them within existing norms, heuristics, beliefs, and preferences. Accordingly, they develop updated understandings of these problems and their potential solutions, guiding policymaking (Beach et al., 2021; Wagner & Ylä-Anttila, 2020). How these informational cues are exchanged, selected, and processed significantly impacts what is learned and the products of learning (Heikkila & Gerlak, 2013; Moyson, 2017; Nowlin, 2021).

However, despite being framed as a key aspect of policy learning, especially over recent years, the role of information processing as a micro foundational dimension of policy learning is often under-theorized (but see Nowlin, 2021). Policy process work on information and information processing developed by scholars of Punctuated Equilibrium Theory (PET) (see Jones & Baumgartner, 2012) and largely independent from policy process work on policy learning. This is even though several decades ago, trailblazers in the field, such as Simon (1997) and Deutsch (1963), among others, pointed to the need for recognizing the cybernetic nature of policymaking and policy learning where the interaction between information and heuristics, biases, and reasoning paradigms substantively shapes policy outcomes. The relative scarcity of theoretical focus on the connections between information processing and policy learning somewhat limits the field's ability to explain the underlying causal mechanisms by which policy actors learn how they do and consequently explain their learning outcomes. This is particularly problematic given the importance of integrating multilevel views of policy learning (micro, meso, and macro) in describing how learning aggregates towards specific outcomes (see Dunlop & Radaelli, 2016; Heikkila & Gerlak, 2013; Zaki et al., 2022).

In this Element, we address this through synthesizing research on learning and information processing to offer insights on how information processing dynamics (i.e., attention to some information signals while ignoring or downplaying others) influences what is or isn't learned and how this contributes to changes in policy learning outcomes. Doing so advances the theoretical debate on learning by further enshrining the micro foundational and fundamental information processing dimension within existing approaches to policy learning and plotting how they influence policy learning processes. This helps provide a more robust baseline for aggregating how learning takes place across multiple levels and thus advances our ability to explain learning outcomes and how learning contributes to policy change (Dunlop & Radaelli, 2017; Moyson et al., 2017). To do so, in Chapter 1, we provide a thorough overview of the foundations of policy learning as a field of study and its development. Then, we showcase how these foundations developed into the contemporary sprawling landscape of policy learning research and its main analytical, theoretical, and conceptual streams (i.e., types, forms, mechanisms, modes of learning, etc.). From there on, we discuss the different ontological approaches to policy learning while dedicating special attention to the role of information processing therein. In Chapter 2, we examine policy-embedded learning as a conceptual background for thinking about the role of information and information processing learning. Then, we discuss the role of belief systems and information processing in policy learning, from which we build a model of policy learning through information processing. Next, in Chapter 3, we build on the model and outline a framework of information processing and policy-embedded learning while empirically applying the framework to the case of United States public opinion about nuclear energy following the Fukushima nuclear disaster. This is followed by a set of brief concluding remarks that connect our model and framework to the conceptual, analytical, and theoretical approaches to learning discussed in this element. This also includes a discussion of the limitations of this approach, most notably how policy change happens outside of policy learning through information processing.

1.2 Policy Learning: history, roots, and branches

Decades ago, the contemporary foundations of learning in public policy and administration research were laid by trailblazers such as Dewey (1946), Deutsch (1963), and Lasswell (1970); Lasswell (1951), among others. This was underpinned by the understanding of the practice of administration and policy as driven by the need to respond to novel, unfolding societal challenges (Kaufman, 1969). This intellectual stream was fuelled by the conception of policymaking (i.e., the conduit of addressing societal problems) as a process of collective "awe" and "puzzlement" (Heclo, 1974). As such, early learning scholarship sought to bring a disengaged society and a somewhat alienated scientific community closer to the puzzle board of societal problems. Hence, the central tenet of the then-blossoming learning movement was a form of stakeholder togetherness rooted in pragmatism and aimed at discovering and pursuing what works while avoiding what doesn't (Dunlop et al., 2018; Sanderson, 2009). At the time, this collective action perspective on learning provided a supplementary view to a dominant "powering-based" understanding of policymaking and policy change and thus introduced learning as a new potent yet less visible explanatory dimension to policy analysis (Hall, 1993; Heclo, 1974; Zaki & Radaelli, 2024). This rather pragmatic solution-oriented foundation explains the explosive growth in policy learning research and practice in the following decades of the contemporary learning movement. Accelerating technological and industrial advances, growing connected ecologies, and increasing complexity gave way to new problems, such as wicked and super-wicked policy issues and crises of different shapes and forms. Such problems were notoriously challenging to define, continuously evolving, complex, stakeholder-dense, politically contested, and divisive (Boin et al., 2020; Head & Alford, 2015; Rittel & Webber, 1973). These features have positioned inclusive, collaborative, and comprehensive learning processes (systematized and organic, individual, and collective) as key to developing solutions (Sanderson, 2009; Zaki, 2024c).

This explains the boom in policy learning as a practice where different actors increasingly engaged in systematic policy learning processes. Hence, starting in the 1970s, we began to witness steady growth in research on policy learning across various domains ranging from the environment and climate change (Gerlak et al., 2018; T. Lee & Meene, 2012; Rietig, 2019), to social welfare, economics, and public health (Crow et al., 2023; Dunlop, 2017; Heclo, 1974; Shi, 2012), crisis governance (Kamkhaji & Radaelli, 2017; Taylor et al., 2022; Trein & Vagionaki, 2022), and beyond. Frameworks such as the Open Method of Coordination (OMC), the European semester, and inter-service consultations within the European Union were set in place to foster (among other objectives) benchmarking, cross-country learning, and multidisciplinary learning. Thematic networks such as the C40 network for climate action, among others, also pursued similar learning-oriented objectives (Borrás & Jacobsson, 2004; Candel et al., 2023; Kerber & Eckardt, 2007; T. Lee & Meene, 2012). Interest in policy learning did not only manifest at the apex of governance architectures but was also omnipresent at the sub-national levels. While we have seen policy learning take place at the European Union level and within another large-scale international policy network, we have also seen it take place at the sub-national levels in provinces, cities, and local communities (Ansell et al., 2017; Busetti & Righettini, 2023; Zaki & Wayenberg, 2023).

The drive for more learning in practice has fuelled the theoretical development of policy learning, contributing to the blossoming of what Dunlop et al. (2018) describe as "the family tree of policy learning." What has started at the roots as a process of collective puzzlement in search for solutions through knowledge utilization (Weiss, 1986), engagement of the public in problem-solving (Dewey, 1946; Lasswell, 1970), development of negotiated solutions (Lindblom, 1959), reasoning of information within government and societal networks (Simon, 1997) grew into various theoretical branches. Some focus on the role of ideas in learning (Ettelt et al., 2012), the migration and mobility of policies through transfer, diffusion, convergence (Dolowitz & Marsh, 2000), information processing (Nowlin, 2021; Zaki et al., 2022), the use of evidence (Sanderson, 2009), and typological views of learning aimed at achieving different types of actor objectives

(Biegelbauer, 2016; Dunlop & Radaelli, 2013). Following the family tree metaphor of Dunlop et al. (2018), these theoretical branches fed into different outcomes or fruits. There, we see different utilization of learning to investigate or shape policy dynamics, for example, design-based approaches to learning (Zaki, 2024b), causal approaches to policy and belief change (Dunlop & Radaelli, 2017; Sabatier & Jenkins-Smith, 1993), and learning as a lens on the policy process itself (Dunlop et al., 2018; Heikkila & Gerlak, 2013). Now that we have briefly mapped the history and development of policy learning as a field of study, we will outline some of its main analytical, theoretical, and ontological approaches.

1.3 Policy Learning: A Synthesis of Key Conceptual, Analytical, and Theoretical Approaches

1.3.1 Key Conceptual Approaches

The increasing recognition of policy learning as a key lens for policy analysis resulted in a flurry of research. While this has enriched our understanding of policy learning, it has also resulted in a vast array of conceptual approaches and theoretical descriptors with overlapping contours, often without arrangement or mapping. Furthermore, the multilevel and implicit nature of policy learning and its normative appeal have made it relatively conceptually ambiguous and challenging to identify. This rendered policy learning something of a "hembig," i.e., a hegemonic yet excessively scoped concept, thus leading to issues of conceptual stretching, ambiguity, and challenges in systematizing and cross-comparing findings. That concoction of issues led Levy (1994) to liken the study of policy learning to sweeping a conceptual minefield. Consistently, recent reviews of learning literature have pointed to the existence of 'conceptual jungles' of overlapping contours where it becomes hard to reach a consensus over one or even a handful of ways to define learning (Borrás, 2011; Gerlak et al., 2018).

This state of affairs stimulated calls for theoretical and conceptual syntheses upon which further field development can be established (Dunlop & Radaelli, 2018b; Goyal & Howlett, 2018; Zaki et al., 2022). Before delving deeper into the information processing aspect of policy learning, it is of utility that we first lay out the field's key conceptual and theoretical approaches. The objective of doing so is twofold. First, providing a succinct synthesis of the sprawling field of policy learning, and second laying a robust foundation on which to map out the dynamics of policy learning and information processing.

We begin with a discussion of what policy learning is. Interest in learning has led to diverse approaches to understanding what it is, often distinguished by their underlying ontological orientations (Dunlop & Radaelli, 2017; Zaki & Radaelli, 2024). These approaches spanned a spectrum of ontological-epistemological perspectives, including mechanistic, positivist, interpretivist, social constructivist, and discursive-institutionalist paradigms (Grin & Loeber, 2007; Oliver & Pemberton, 2004). For instance, the constructivist interpretivist approach emphasizes learning as a social process involving the negotiation of problem definitions, communication, interpretation, understanding, and assessment of solutions (Grin & Van De Graaf, 1996). Conversely, other perspectives examine how actors process new information, primarily through pre-existing coherent belief systems (Dudley et al., 2000; Sabatier & Jenkins-Smith, 1993).

This diverse ontological-epistemological foundation has also spilled over to the field's conceptual landscape, boasting many definitions and conceptual approaches. According to a recent review by Zaki et al. (2022), there are over 30 distinct definitions of learning in public policy literature. The most prominent definition of learning follows the Advocacy Coalition Framework (ACF) (Sabatier & Jenkins-Smith, 1993), viewing learning as enduring changes in thoughts, beliefs, or behavioral intentions regarding policy problems. Other scholars, such as Hall (1993), view learning as a deliberate attempt to adjust the goals or techniques of policy in response to experience and new information. On the other hand, we also have definitions that leverage policy learning as an umbrella concept for several similar phenomena encompassing the general mobility of ideas covering varieties of emulation, transfer, and lesson drawing where knowledge about policies, solutions, and administrative arrangements flow across time and/or space (Giest, 2017). Furthermore, there are also approaches that view policy learning as a general increase or update of knowledge about policy problems (Corbett et al., 2018), rational reflections on public policy using acquired knowledge and information (Newman & Bird, 2017), or the acquisition, translation, and dissemination of information among actors with diverse knowledge bases (Heikkila & Gerlak, 2013). This is in addition to conceptual approaches that view learning as a decision-making paradigm or an approach to policymaking where policy decisions are made based on knowledge, past experiences, and future expectations (Karlsen & Larrea, 2017). Put together, conceptual approaches to learning can be viewed across four main perspectives, including mechanism and process-oriented, paradigmatic, outcome-oriented, and ideational. The perspectives, how they are conceptualized, and some representative citations are shown in Table 1.1.

Table 1.1: A Non-Exhaustive Synthesis of Conceptual Approaches to Policy Learning

Perspective	Conceptualization
Mechanism and process-oriented: Focused on the processes and mechanisms underlying how learning takes place	The acquisition, translation, and dissemination of information among actors with diverse knowledge bases (Heikkila & Gerlak, 2013). The transmission of policy knowledge between political actors (Legrand, 2012). A deliberate and critical reflection on prior knowledge (Rietig & Perkins, 2018).
Paradigmatic: Learning as a decisional approach or a policymaking style	An approach to policymaking where policy decisions are made based on knowledge and past experiences and future expectations (Karlsen & Larrea, 2017). Learning as a a deliberate attempt to adjust the goals or techniques of policy in response to experience and new information (Hall, 1993).
Outcome oriented: Focused on the products of learning such as cognitive updates and changes in policy positions	Enduring changes in thoughts, beliefs, or behavioral intentions regarding policy problems (Sabatier & Jenkins-Smith, 1993). The general increase in knowledge about policies (Corbett et al., 2018).
Ideational: Learning as the mobility of ideas and policies across time and space	A process whereby knowledge about policy in one jurisdiction is acquired and utilized in decisions regarding the development of policies in another (Newman & Bird, 2017). Drawing lessons from experiences, and using knowledge to inform policy action (Motta, 2018).

One way of navigating and systematizing many definitions in learning literature is by looking at the background and systematized levels, where we can understand what a concept is inherently really about (see Adcock & Collier, 2001). Policy learning can be broadly understood as an iterative problem-oriented process where policy actors seek and process information about policy issues, aiming to update their understanding of how they can be solved (Zaki et al., 2022).

This can lead to different outcomes, including *cognitive*, i.e., developing or updating ideas and beliefs about policy problems and potential solutions. These products can sometimes lead to *behavioral* outcomes, i.e., implementing political strategies and adjusting or changing policies to address problems (Heikkila & Gerlak, 2013).

Adopting this overarching and fundamental information processing-based view of learning where precise analytical dimensions exist (i.e., information and knowledge, actors, learning structures, policy problems, and contexts) can help identify and analyze learning across different conceptual approaches. For example, policy transfer and lesson drawing could be broadly understood as involving the mobility of policy ideas from one jurisdiction or experience to another (D. P. Dolowitz & Marsh, 2000; Rose, 1991; Volden et al., 2008). The information processing background approach to learning can help structure the analysis of lesson drawing and policy transfer by pinning down the key dimensions of such processes. This can be done by analyzing the sources of information used, actors, and the cognitive and governance structures involved.

With the conceptual foundations and streams briefly presented, we then focus on key theoretical perspectives in policy learning. Namely, we focus on analytical perspectives and theoretical descriptors such as forms, mechanisms, modes, and types of learning.

1.3.2 Key Analytical Approaches: Learning as Levels of Analysis

As discussed, several approaches have emerged to describe the growing literature on policy learning and capture diverse learning phenomena. The very concept of learning can provide different analytical perspectives on the policy process. In addition, policy learning is positioned to explain different policy and organizational behaviors across multiple levels of analysis: micro, meso, and macro. Table 1.2 provides descriptions and examples of learning at the micro, meso, and macro levels, as well as multidimensional and non-linear learning.

Table 1.2: Learning as Levels of Analysis

Levels of Analysis	**Examples**
Micro-level studies: Focus on individuals and cognitive foundations	Learning as a cognitive micro foundational process (Beach et al., 2021), behavioral perspectives moderating learning (Zito, 2009), the role of individual beliefs and information flows in learning (Moyson, 2017; Nowlin, 2021; Pattison, 2018), biases and heuristics (Heikkila et al., 2024; Malkamäki et al., 2021; Wagner & Ylä-Anttila, 2020).
Meso-level studies: Focus on organizational units, collectives and networks	Learning as interactions with information and beliefs within and across advocacy coalitions (Sabatier & Jenkins-Smith, 1993; Weible, 2008), learning as interaction within and with expert communities (Haas, 1992), learning within networks (Howlett et al., 2017), learning within organizations (Borrás, 2011; Zaki & Dupont, 2024; Zito, 2009).
Macro-level studies: Focus on systems	Learning and deliberation in transnational policy issues and crises (Dunlop et al., 2020), Collective puzzlement and the role of ideas and paradigms (Hall, 1993; Heclo, 1974), learning as an inter-system transfer and diffusion of ideas (D. P. Dolowitz & Marsh, 2000; Marsh & Sharman, 2009), learning and systemic convergence towards certain policy ideas (Bennett, 1991).
Multidimensional and non-linear learning: Focus on cross-level interactions	Learning as a multidimensional and interactive process between micro, macro, and meso levels (Dunlop & Radaelli, 2017), the interaction of learning types over time, learning type shifts and interactions over time (Biegelbauer, 2016; B. Zaki et al., 2024).

With the different key analytical perspectives in policy learning literature established, next we turn our attention to the various theoretical descriptors of policy learning.

1.3.3 Key Theoretical Perspectives: Disentangling Theoretical Descriptors of Learning

How can we theoretically describe or unpack policy learning with such a rich and diverse epistemological-ontological foundation contributing to conceptual pluralism? As the field grew, theoretical learning descriptors emerged based on empirical analyses. Recent reviews by Gerlak et al. (2018) and Zaki et al. (2022) take stock of over 60 different theoretical descriptors of learning, typically aggregated under the label of "types of learning" and often without clear definitions or dimensions that enable identification or differentiation. For conceptual and theoretical clarity, we disentangle these theoretical perspectives, pointing to types of learning, modes of learning, forms of learning, mechanisms of learning, and degrees of learning, as shown in Table 1.3.

Table 1.3: A Non-Exhaustive Account of Theoretical Descriptors of Learning

Types of Learning	Modes of Learning	Mechanisms of Learning	Forms of Learning	Degrees of Learning
Instrumental Social Political Policy-oriented Organizational/Institutional	Epistemic Hierarchical Reflexive Bargaining-oriented	Contingent Inferential	Lesson drawing Policy Transfer and imitation	Single and double loop Triple loop Quadruple loop

Types of Learning: The most commonly used theoretical descriptor of learning is type, where a learning type refers to a learning process's intended outcome or orientation and points to the object of learning or what is being learned about. Some of the most prominent and clearly defined learning types include *instrumental* learning, which is focused on updated understandings of policy instrument calibrations and viability (Lee & Meene, 2012; May, 1992), and *social* learning, which focuses on the evolving social construction of policy issues through discursive and reflexive processes (Hall, 1993). Another widely studied learning type is *political* learning, which is concerned with policy actors updating their understanding of strategies and techniques required to advance the political viability of their policy positions, preferred solutions, and interests (Heclo, 1974; Zaki & Dupont, 2024). An additional type of policy learning is *policy-oriented* learning, where learning is understood as cognitive or behavioral changes as a result of updating policy beliefs (Jenkins-Smith & Sabatier, 1993; Sabatier, 1987, 1988). Additionally, policy-oriented learning is limited to learning about the policy issue, including beliefs about the nature of the problem, its causes and consequences, who is harmed, and how the problem could best be addressed, among other beliefs. Finally, there is *organizational* learning, which focuses on updating organizational structures, norms, and procedures (Huber, 1991; Zito, 2009).[1]

Modes of Learning: Apart from learning types, there are also modes of learning that refer to how learning interactions occur. This includes *epistemic* learning from socially endorsed subject matter experts, *hierarchical* learning within formal organizational structures using top-down communication and the dissemination of knowledge, and *bargaining-oriented* learning involving negotiation, trade-offs, and strategic interactions between policy actors. Lastly, *reflexive* learning involves utilizing knowledge to deepen discussions and facilitate arguments (see Dunlop & Radaelli, 2013).

Mechanisms of Learning: Next are mechanisms of learning, which involve the processes by which learning materializes, pointing to the series of interactions by which learning occurs. There are two main learning mechanisms: inferential and contingent. In *inferential* learning, policy actors face a problem and then resort to critical and deliberate reflection on priors, leading to an updated understanding of policy problems, viable solutions, and potential policy change (Hall,

1 There are also other learning types not shown in Table 1.3, including strategic learning, focused on learning about policies and procedures that fulfill compliance mandates or externally generated objectives (Common & Gheorghe, 2019), and managerial learning, focused on updated understandings of policy implementation arrangements (Biegelbauer, 2016).

1993). With *contingent* learning, on the other hand, surprise (systemic shocks or crises) trumps critical reflection on priors (Kamkhaji & Radaelli, 2017). In this case, policy actors undertake rapid behavioral changes through associative mechanisms to avoid imminent harm. This means that policymakers quickly adapt their behaviors and policies when they encounter unexpected situations by drawing on a reservoir of cue-outcome associations. Once the crisis shock dissipates, critical reflection on priors and further policy adjustments or lock-ins occur. Accordingly, policy changes occur before learning.

Forms of Learning: The forms of learning concern the actualization or the process identity of learning, its shape, structure, and form. For example, learning can occur through *lesson drawing*, where policy actors learn from their experiences across jurisdictions to address similar issues in their contexts by critically analyzing and adapting successful policies or programs to fit local needs (Rose, 1991). It can also occur through *policy transfer*, where policies, administrative arrangements, institutions, and ideas travel from one political setting to another. That process can include voluntary and coercive transfers, where policies are adopted by choice or due to external pressures (D. Dolowitz & Marsh, 1996).

Degrees of Learning: Finally, the degrees of learning refer to the depth or extent of learning. Here, the literature offers a four-scale gradient that includes single, double, triple, and quadruple-loop learning. *Single-loop* learning focuses on identifying adjustments within existing policy instruments without questioning the underlying assumptions or objectives and mostly attempting to achieve better outcomes within the current policy framework. *Double-loop* learning delves deeper by reflecting on and examining the underlying policy goals, assumptions, and strategies. This process involves potential changes in fundamental principles and approaches to underlying policies (Argyris & Schon, 1978). *Triple-loop* learning consists of reflecting on the learning process itself, mainly by challenging existing rules and norms, leading to changes in how knowledge is acquired and used in the learning process (Tosey et al., 2012). The fourth and most profound level of learning is *quadruple-loop* learning. This occurs when broader social, political, and contextual factors are incorporated into the learning process. This leads to continuous adaptations of norms, values, procedures, and learning process structures (S. Lee et al., 2020). It is important to note here that these degrees of learning are often mapped against orders of change where single, double, and triple-loop learning are associated with first, second, and third-order change, respectively (see Hall, 1993; Heclo, 1974; B. Zaki et al., 2024).

The above review of the vibrant policy learning literature above shows a landscape sprawling with a wide range of ontologies and conceptual and theoretical approaches that continuously grow based on empirical analyses across a wide range of policy domains. While this literature continues to provide insights into policy processes and outcomes, the persistent issue of how learning occurs at the micro foundational level and aggregates across the micro, meso, and macro levels, from individuals to collectives, remains scarcely theorized and challenging to address (Dunlop & Radaelli, 2017; Jones, 2017). In the next chapter of this Element, we turn our attention to this issue and outline a coherent theoretical approach to understanding the micro-foundations of learning and its aggregation across multiple levels of analysis. At the root of the policy learning model we develop is how information is processed through filters derived from belief systems. We argue that the learning through information processing approach is present at the micro, meso, and macro levels with the same underlying dynamics at each level.

2. Information Processing and Belief Systems

2.1 Introduction

At the most fundamental level, information constitutes the fuel of individual and collective behavior related to policymaking. Hence, recent policy process scholarship has sought to combine the policy-oriented type of learning from the Advocacy Coalition Framework (ACF) with the information processing perspective of Punctuated Equilibrium Theory (PET) (Nowlin, 2021, 2024). However, the broader policy learning literature has had an inherent and essential focus on information that helped advance it to the ranks of policy process ontologies, enabling the learning lens to generate meaningful insights into policymaking processes by deciphering why, when, and how policy actors engage with information to learn about policy problems (Kamkhaji, 2022; Zaki et al., 2022; Zaki, 2024b). So, how do information and knowledge primarily feature in policy learning research?

Policy learning literature engages with the role of information and knowledge in learning; however, this is predominantly done at the meso and macro levels. For example, at the meso-level, there is a longstanding literature exploring the role of information and knowledge in public organizations' learning processes (Arnold, 2014; Jenkins-Smith, 1988; S. Lee et al., 2020), advisory groups and different sub-organizational units (Candel et al., 2023; Mavrot et al., 2024; Nair & Garg, 2024; Zaki & Wayenberg, 2021), and within and across coalitions operating in policy subsystems (Jenkins-Smith & Sabatier, 1993; Nohrstedt et al., 2023; Sabatier & Jenkins-Smith, 1993). Similarly, there is substantive research on the role of information and knowledge at the macro and systemic levels, which involves looking at governance systems and societal paradigmatic transformation induced by the introduction of new information and knowledge, whether through instrumental, reflexive, or social learning (Challies et al., 2017; Hall, 1993; Heclo, 1974; B. Zaki et al., 2024). These contributions have helped develop nuanced theoretical understandings of the role of information and knowledge in policy learning (and thus policy change). However, on the micro-foundational level, research remains somewhat limited overall. Here, we find remarkable yet sparse contributions, mainly focusing on exploring the influence of reasoning paradigms and heuristics on learning (Beach et al., 2021; Malkamäki et al., 2021; Wagner & Ylä-Anttila, 2020). Hence, a granular and micro-foundational understanding of how information is processed in the learning process, leading to cognitive and behavioral outcomes, remains scarcely theorized. This limits our ability to coherently analyze how learning occurs at the individual level and aggregates to the collective and then trace how this learning ties into and drives varieties of change.

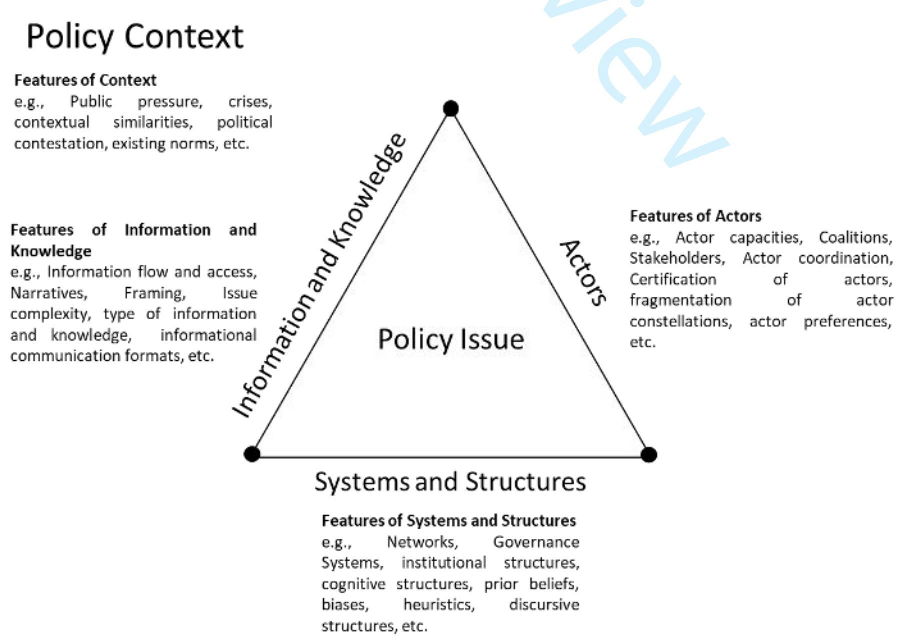
In this chapter, we integrate belief systems and policy-oriented learning from the ACF, information processing from PET, and the broader policy learning literature to develop a systematic and theoretically coherent approach to understanding the role of information and information processing in policy learning and policy change. In the next section, we establish a conceptual backdrop upon which an information-processing approach to learning is built. Next, we discuss belief systems and learning, including connections between beliefs, issue definitions, and policy preferences. Then, we discuss information and information processing. Finally, we develop a model for analyzing policy learning as an outcome of information processing.

2.2 Policy Learning: A Conceptual Backdrop for an Information-Processing Approach

Policy learning has often been considered a puzzling and analytically elusive phenomenon. Hence, as noted in Chapter 1, different conceptual approaches to viewing or unpacking learning have been developed. While these approaches successfully capture a wide range of perspectives on learning, in many cases, they do not offer the precise conceptual dimensions that enable robust and systematic analyses. Along with the fundamentally elusive nature of learning and its normative appeal as a concept, this led to learning research being labelled as a conceptual minefield, an overly crowded landscape with a dizzying array of definitions, or even as a metaphor rather than a coherent analytical lens (see for example Borrás, 2011; Goyal & Howlett, 2019; Levy, 1994).

The first step to building a coherent theoretical or analytically viable approach to analyzing learning is to start from a clear conceptual foundation with coherent ontological-epistemological assumptions that drive our analytical approach (Zaki & Radaelli, 2024). With a focus on the information processing perspective on learning, we draw on a cybernetic and mechanistic background conceptualization of *policy-embedded learning*. With policy-embedded learning, learning can be understood as "The circulation and consumption of policy issue-related information and knowledge among policy actors embedded within a policy system within multilevel structures in a certain policymaking context" (Zaki et al., 2022, p. 22). The policy-embedded learning approach is shown in Figure 2.1.

Figure 2.1: Policy-Embedded Learning



The policy-embedded approach to learning was developed to address some of the field's key conceptual and analytical challenges by providing a background conceptualization of the policy learning process that comports with (and supplements) existing approaches. This is done by serving three key purposes. First, identifying key fundamental and omnipresent analytical dimensions valid across different conceptual and field-specific approaches to learning (i.e., policy actors, information and knowledge, systemic and cognitive structures, and policymaking contexts). Second, these fundamental dimensions should be ensured to apply to different analytical levels (micro, meso, and macro). Thus, they could be aggregated to trace the relationship between policy learning and its outcomes more comprehensively. Third, and relatedly, it provides the ability to address the key limitation of several existing approaches to learning in the public policy literature, which has been largely borrowed from cognitive functional psychology, where learning is understood in terms of the impact of experience on behavior or as materializing as enduring changes in the mechanisms of behavior (Lachman, 1997; Ormrod, 2019). The cognitive approach, while having its benefits, endures fundamental issues in providing the ability to analytically separate the learning process from its outcomes, mainly due to the inherently obscured causal pathways between experience, behavioral change, and learning on the one hand and the latencies between cues and perceived outcomes on the other (De Houwer, 2011; De Houwer et al., 2013). These issues become exacerbated once the same approach is ported over to a messy and politicized context as that of policymaking (Zaki et al., 2022).

With the three elements discussed above in place, the policy-embedded framework to learning places a premium on the role of information processing in shaping learning processes and outcomes while accounting for the interplay between policymaking contexts, as well as different levels of structures surrounding the learning process, including micro-foundational such as prior beliefs, biases, heuristics and sense-making frames, as well as more systemic structures such as institutional setups, rules, and governance structures. The emphasis on information and knowledge in this approach, as well as in the broader policy learning literature, can be attributed to its core role in shaping policy actors' understanding of policy problems and, thus, their patterns of cognitive update and changes in policy behavior. For example, literature shows that various features of the information fed into the learning process (e.g., its disciplinary nature, complexity, accessibility, robustness, and credibility) and the cognitive frameworks through which it is processed or filtered (e.g., biases, heuristics, prior beliefs, and policy preferences) significantly influence learning outcomes (Malkamäki et al., 2021; Moyson, 2017, 2018; Nowlin, 2021, 2024; Wagner & Ylä-Anttila, 2020; Zaki et al., 2022).

Having established a conceptual approach to information-processing-based policy learning and emphasizing the centrality of information and information processing therein, we next develop a model for policy learning and information processing, which serves as the engine that drives learning within the policy-embedded learning framework. The model of learning through information processing draws from Nowlin (2021), with an emphasis on the ACF's notions of policy-oriented learning and beliefs and work on information processing derived from PET.

2.3 Belief Systems and Learning

The ACF, with its focus on policy-oriented learning as a key driver of policy change, is the policy process theory with the most developed understanding of the importance of policy learning. Policy-oriented learning within the ACF is a cognitive process based on policy beliefs and belief change, as well as how beliefs inform behavior. It is defined as "enduring alterations of thought or behavioral intentions that result from experience, and which are concerned with the attainment or revision of the precepts of the belief systems of individuals or of collectives" (Jenkins-Smith &

Sabatier, 1993, p. 42). Hence, learning here involves changes in thought and behavior that result from changes in the belief systems of individuals or collectives. This definition pivots on a cognitive process that produces belief and behavioral change while also recognizing that learning occurs among individuals and collectives (also see Gerlak & Heikkila, 2011; Heikkila & Gerlak, 2013).

The ACF posits a hierarchical, three-tiered belief system that includes deep core beliefs, policy core beliefs, and secondary beliefs. Within the belief hierarchy, beliefs move from abstract foundational beliefs – deep core beliefs – to beliefs specific to a policy issue or subsystem – policy core beliefs – to secondary beliefs, which focus on particular means to achieve policy goals. Deep core beliefs inform and constrain policy core beliefs, and deep core and policy core beliefs shape secondary beliefs.

Advocacy coalitions are formed around shared policy core beliefs. Policy core beliefs are relevant to a particular policy issue, domain, or subsystem. Additionally, policy core beliefs contain a normative element that "may reflect basic orientation and value priorities for the policy subsystem," such as "whose welfare … is of greatest concern"; and empirical considerations that involve "overall assessments of the seriousness of the problem, basic causes of the problem, and preferred solutions for the addressing the problem" (Jenkins-Smith et al., 2018, pp. 140–141). The components of policy core beliefs reflect the fact that most policy issues are complex and contain multiple attributes. The various attributes of complex policy issues can be considered dimensions in a multi-dimensional issue space (Jochim & Jones, 2013; Jones & Baumgartner, 2005b).

Policy core beliefs regard complex policy issues that contain multiple dimensions; however, policy actors and institutions are not able to consider all of the possibly relevant dimensions and, therefore, only focus on a few dimensions of the issue (Jones, 2001, 2017; Jones

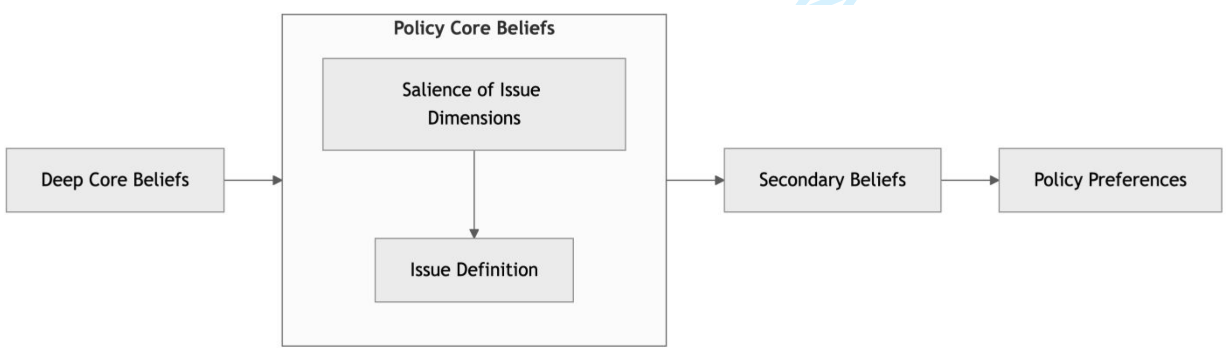
& Baumgartner, 2005a). Apart from being able to only attend to a limited number of dimensions, policy actors also view some dimensions of the issue as more salient or important than others. As a result, issues can be understood or defined in different ways through various combinations of issue dimensions and the salience of those dimensions. For example, assume the issue of gun violence in the US has two dimensions, including the prevalence and availability of guns as well as the mental health of those who commit violent acts using guns. For some, the prevalence of guns is the salient, defining dimension, but for others, the mental health dimension is the important dimension. As a result, the issue of gun violence is defined in two separate ways. Previous work, building on notions of problem definitions, policy images, and frames, notes that the aggregation of the various dimensions weighted by the salience of each dimension constitutes an issue definition (Nowlin, 2016). In sum, Nowlin (2016) posited that any policy issue, I, contains D_{In} possible dimensions, and each dimension is weighted $D_{\text{An}}(\omega)$ with regard to salience. Therefore, issue definitions are the aggregate of various dimensions weighted by salience. We argue that such issue definitions are the nucleus of policy core beliefs.

Issue definitions are the sum of the various dimensions of the issue weighted by the salience and form the foundation of policy core beliefs. Policy core beliefs are driven by deep core beliefs. Therefore, deep core beliefs influence the salience weights given to the various issue dimensions. With the gun violence example, the deep core beliefs of conservatism and individualism coupled with normative policy core beliefs that value the rights of gun owners would likely lead to seeing the mental health of individuals, rather than the availability of guns, as the defining dimension of gun violence.

It is also important to distinguish between policy beliefs and policy preferences. As noted by Plümer (2024), policy preferences “refer to the goals of actors within a specific policymaking context" (9). Some argue that policy preferences are policy core beliefs associated with preferred policy approaches and constitute policy core policy preferences (Jenkins-Smith et al., 2018). While policy beliefs and policy preferences are both cognitive learning products, beliefs exist a priori and inform preferences in the same way beliefs inform attitudes (Fishbein & Ajzen, 2011; Nowlin, 2019). Beliefs can inform the perceived effectiveness of a policy approach, whereas preferences, like attitudes, are stated as "I support" or "I oppose" that policy approach (Moyson, 2017; Plümer, 2024). As noted, the hierarchical nature of belief systems means that beliefs range from abstract to particular, with the more abstract beliefs influencing the more particular beliefs. Therefore, secondary beliefs are the beliefs closest to policy preferences. For example, a secondary belief that a carbon tax is an efficient and effective way to address climate change would translate into support for a carbon tax, i.e., a policy preference.

Figure 2.2 illustrates our formulation of the relationships between policy beliefs, issue definitions, and policy preferences.

Figure 2.2: Policy Beliefs, Issue Definitions, and Preferences



As shown in Figure 2.2, deep core beliefs shape policy core beliefs, which consist of the issue definition or the aggregated salience of the various issue dimensions. Policy core beliefs inform secondary beliefs, which in turn inform policy preferences. While learning as belief updating could occur for any belief, they become less likely to change as they become increasingly applicable to multiple policy issues. It is rare that deep core beliefs such as political ideology or cultural worldviews change, whereas narrowly focused secondary beliefs are the most likely to change. Policy core beliefs are more likely to change than deep core beliefs but less likely than secondary beliefs. Similar to secondary beliefs, policy preferences are more likely to change than policy core or deep core beliefs.

Policy-oriented learning is associated with belief system *changes* within a subsystem. However, ACF scholarship has also noted that belief reinforcement can be a type of or an outcome of policy-oriented learning (Nohrstedt et al., 2023; Pattison, 2018). Indeed, belief reinforcement is likely the most common type of policy-oriented learning (Weible et al., 2023). Recent work has proposed that policy learning as change and reinforcement can be explained when policy beliefs are understood as a probability distribution with a central tendency (the belief) and variance (uncertainty about the belief) (Nowlin, 2024). With this approach, learning as belief change occurs when there is a shift in the central tendency, and learning as belief reinforcement occurs when there is a reduction in variance. Using this approach, policy-oriented learning outcomes include no change in beliefs (i.e., no learning), belief reinforcement (i.e., a reduction in uncertainty), belief change (i.e., learning, or a significant shift in the central tendency), or belief change and reinforcement (i.e., a shift in beliefs and a reduction of uncertainty) (Nowlin, 2024). The likelihood of belief change is a function of the strength of the belief with firmly held beliefs with low uncertainty less likely to change than less firmly held or less certain beliefs (Jenkins-Smith et al., 2018; Nowlin, 2021).

Additionally, policy-oriented learning occurs both within coalitions and between coalitions. Learning within coalitions is relatively straightforward given the nature of their shared beliefs; however, learning between coalitions is, in part, a function of the collective context in which information is processed, and the issues are discussed. According to ACF scholarship, learning across coalitions is more likely when a) there is an intermediate level of conflict because in such contexts, coalitions have the resources to engage and the core beliefs of both coalitions are not at stake; b) issues that are analytically tractable and high-quality data is available; and c) a forum that encourages participation of representatives of the various coalitions and professional norms of engagement (Jenkins-Smith & Sabatier, 1993).

Another important context for policy-oriented learning is the nature of the policy subsystem in which policy actors engage. Subsystems are the key units of analysis within the ACF and can vary in several key dimensions. Weible (2008) noted that subsystems can be unitary, collaborative, or adversarial. Unity subsystems involve one dominant coalition and a single understanding of the policy issue. Policy-oriented learning in unitary subsystems happens within the dominant coalition and is likely belief reinforcement (Weible, 2008). Additionally, in unitary subsystems, policy-oriented learning as belief change is likely impeded by the status-quo bias of the dominant coalition and the negative feedback exerted regarding new information (Jones & Baumgartner, 2005b). However, collaborative subsystems involve cooperative coalitions with a shared understanding of the policy issue and learning as belief change and belief reinforcement may occur within and across coalitions. Finally, adversarial subsystems include two or more coalitions competing for policy influence with a debated understanding of the policy issue. Since beliefs are contested in adversarial subsystems, belief reinforcement within coalitions is the most likely policy-oriented learning in an adversarial subsystem.

Policy-oriented learning occurs within and across individuals and collectives as part of a collective learning process. More broadly, collective learning can be understood as consisting of two parts, including "(1) a 'collective process,' which may include acquiring new knowledge through diverse actions (e.g., trial and error), assessing information and disseminating new knowledge or opportunities across individuals in a collective, and (2) 'collective products' that emerge from the process, such as new shared ideas, strategies, rules, or policies (Gerlak & Heikkila (2011), 623). The first part, the process of learning, involves acquiring knowledge, whereas the second part, learning products, involves the outcomes associated with the process of learning.

A key factor in the learning process is information, including its availability and how it is processed in a collective context. In the next section, we focus on the role of information and information processing in learning.

2.4 Information Processing

The work on information processing in policy process research has been developed by scholars using the Punctuated Equilibrium Theory (PET) of the policy process (Baumgartner et al., 2023; Baumgartner & Jones, 1993, 2015; Jones & Baumgartner, 2005b, 2012). PET explains policy processes that involve periods of stability and incremental change coupled with periods of significant, rapid, and punctuated change. In brief, PET posits that stability and incremental change are a product of issues being contained within a subsystem, and punctuations occur as issues move between a subsystem and the macro-institutions of policymaking through positive feedback, changing policy images, shifting policy venues and/or new policy actors becoming engaged. As such, this has helped PET expand into a theory of information processing and attention in political systems.

Information and information processing for policymaking include a sender of the information and a receiver of the information. In general, information for policymaking is imperfect, costly, and asymmetric (Baumgartner & Jones, 2015). The imperfection of information relates to that because of knowledge and cognitive limitations, individuals can't acquire all the information required to fully understand all the relevant dimensions and possible outcomes associated with a decision. As a result, individuals "satisfice" or acquire enough information so that they are satisfied that they have enough knowledge to make a decision (Simon, 1955, 1997). Additionally, information is costly for both the sender and the receiver. For the sender, the costs of information involve its acquisition, and for the receiver, the costs are associated with processing. Often, the sender has developed, at great cost, some expertise about the policy issue that other policy actors do not have. As a result, information for policymaking is asymmetric since equal knowledge about an issue is not shared among all policy actors (Krehbiel, 1991).

Another important component of information for policymaking is its supply. Information may be private or public, and keeping or sharing information is a strategic decision for policy actors. For example, a policy actor with expertise about a policy issue may keep some information private so that they can influence policy decisions towards their preferences (e.g., Niskanen, 1971). However, if a policy actor is not providing information, other actors rush to fill in the gap. Therefore, information in policymaking tends to have an abundant supply because policy actors find it in their interest to be public about their knowledge in an effort to shape how policy issues are defined (Baumgartner & Jones, 2015; Workman et al., 2009).

Given the incentives for policy actors to share information, information is not a scarce resource for policymaking. Indeed, policy actors and policymaking institutions are bombarded with information from various sources. As a result, information and the prioritization and processing of information are key to the policy process (Workman et al., 2009). In the information processing approach, information is understood as "signals of change" that emanate from the broader societal environment, and information processing is the "collecting, assembling, interpreting, and prioritizing" of those signals (Jones & Baumgartner, 2005b, p. 7). Information signals that flow into the policymaking system vary but can be categorized as belonging to one of the streams identified by the Multiple Streams Framework (Herweg et al., 2023; Kingdon, 1984), including information about the problem, possible policy approaches or solutions, and the politics surrounding the issue (Nowlin, 2019). Information processing involves processing these signals; however, given that the supply of information possibly relevant for policymaking is large, neither individuals nor institutions can adequately process the vast number of information signals they receive.

Attention to information resulting from the inability of actors or institutions to attend to all information becomes a key limiting factor for how information is translated into policy learning and, ultimately, to policy change or stability. Individuals and institutions have limited capacity to process all relevant and available information. In addition, policymaking institutions impose costs that add friction to the process of translating information into policy action (Jones & Baumgartner, 2005b). These dynamics lead to disproportionate information processing, where some signals are inadvertently overemphasized while others are ignored or de-emphasized (Jones & Baumgartner, 2005b; Jones & Thomas, 2012; Walgrave & Dejaeghere, 2016). Disproportionate information processing limits the policy learning that could occur from potentially relevant information.

Several factors limit the ability of individuals to process all relevant information, with the most notable being the bounded rationality of individuals (Jones, 2002). In addition, attention to some information signals rather than others is driven by policy actors' beliefs, biases, and motivations. Actors motivated by accuracy would likely process information differently than those motivated to reach or maintain a specific conclusion or predetermined policy choices (Druckman & McGrath, 2019; Kunda, 1990; Zaki & Wayenberg, 2021). Along similar lines, prior beliefs can shape how leading policy actors process information to weight information based on its accordance with policy beliefs (Nowlin, 2021).

The nature of political institutions – the rules that govern collective action and shape collective outcomes – also constrain information processing by imposing costs on collective choices, including decision costs and transaction costs (Jones & Baumgartner, 2005b). Decision costs are the costs associated with decision-making or coming to an agreement. These costs involve bargaining and persuasion, but most notable are the decision costs imposed by the structure of policymaking institutions and the level of agreement that is necessary to decide. Systems and structures that have separation of powers, shared authority spread across multiple institutions, and several veto points impose high decision costs (Tsebelis, 2002). Decision costs also include the transaction costs associated with policy legitimation (e.g., voting on a bill or promulgating a rule) that are incurred once a decision is made (Jones & Baumgartner, 2005b).

Two other types of costs are also present: information costs and cognitive costs. Information costs involve costs associated with the search for relevant information for decision-making, and cognitive costs are related to the inability of actors and institutions to process all of the relevant information (Jones & Baumgartner, 2005b). These costs are incurred once it becomes apparent that a decision or a policy change needs to be made. As noted, actors and institutions have more information about problems, policies, and politics than they are able to process, yet as attention to a problem arises, policy actors search for information about the problem. For example, the fact that the concentration of greenhouse gases is rising in the atmosphere due to human activity is an information signal within the “problem” information stream (Herweg et al., 2023; Kingdon, 1984; Nowlin, 2019). Yet, that information is largely unnoticed until the issue of climate change starts to receive attention from a broad array of policy actors and institutions and becomes salient (i.e., placed on the policymaking agenda) (Jones & Baumgartner, 2005a, 2005b; see Kingdon, 1984). Then, search and cognitive costs limit which additional information signals are processed prior to a decision or policy change.

An additional element of search is that information search tends to lead to the finding of problems that need to be addressed, therefore making policy action more likely. Baumgartner & Jones (2015) termed this the paradox of search. Additionally, the paradox of search also entails a trade-off between the diversity of information sources, which leads to an improved understanding of the problem and control over the issue and what actions should be taken. Therefore, policy actors may seek to limit information search to maintain control in an effort to avoid blame for a problem or to curtail government actions for ideological reasons. For example, at a campaign rally in June 2020, President Trump, discussing COVID testing, noted, “When you do testing to that extent, you’re going to find more people, you’re going to find more cases … So I said to my people, ‘Slow the testing down, please.’ They test, and they test” (Freking, 2020). In this example, in Trump’s view, it is likely that increased COVID-19 testing would lead to a loss of control over the issue and to an increase in demand for a government response. However, limiting information search and attempting to control an issue hampers the possibility of learning, though that may be the intent.

Information processing for decision-making, at both the individual and collective level, involves four stages: a recognition stage, a characterization stage, an alternative stage, and a choice stage (Jones & Baumgartner, 2005b). The recognition stage involves allocating attention to particular policy problems and information signals. The recognition stage translates to agenda-setting at the collective level. Information must first be acquired and attended to for learning through information processing. The next stage is the characterization stage, where the various attributes or dimensions of the problem are considered. As noted, most policy issues are complex and can be understood and defined in various ways. The characterization stage involves policy actors assembling and weighing the issue's various dimensions to develop the definition (Nowlin, 2016).

Following the characterization stage is the alternative stage, where policy alternatives are developed. As with the characterization stage, the alternative stage is multidimensional, and several possible policy alternatives are available. The understanding and definition of the problem constitute a multidimensional problem space, and the policy alternatives are a multidimensional solution space (Baumgartner & Jones, 2015). Often, the dimensions of the issues weighted heavily in the construction of the problem space imply particular approaches in the solution space. For example, if the problem of gun violence is defined as mental health, the policy alternatives would focus on addressing mental health issues. Different types of information are needed to construct the problem space versus the solution space. When considering which problems to prioritize, it is best to get information from a diverse array of sources to develop a better understanding of the problem. This information type is called *entropic* (Baumgartner & Jones, 2015). Institutional structures that are decentralized can facilitate this type of learning (Epp, 2018; Heikkila & Gerlak, 2013). However, when considering how to address problems, actors rely on *expert* information. Additionally, the problem and solution spaces can evolve independently, much like the problem and policy streams (Herweg et al., 2023; Kingdon, 1984). The final stage is the choice stage, where a policy alternative from the solution space is selected for support by an individual or selected as a policy by a policymaking institution.

These stages parallel the process of collective learning, which involves the acquisition, translation, and dissemination of information (Heikkila & Gerlak, 2013). The acquisition stage involves individuals acquiring information, which may happen through experience, observation, search, and/or deliberation (Heikkila & Gerlak, 2013). Next, information is translated or interpreted through concerted effort and analysis or through unintentional processes that are influenced by framing or the heuristics of the individual (Heikkila et al., 2024; Heikkila & Gerlak, 2013). The final part of the learning process is dissemination, where individuals share knowledge within a collective. Shared collective routines, communication practices, and deliberation may facilitate the dissemination of information. Additionally, Heikkila & Gerlak (2013) note that dissemination may inform the agenda-setting process and can, therefore, play into the recognition stage.

Intertwining policy-embedded learning, policy-oriented learning, collective learning, and information processing, we argue learning can be understood as a process where policy actors (i) seek and (or) acquire imperfect and costly policy-related information, (ii) process that information through individuals and institutions that are influenced by cognitive structures, biases, and heuristics, and (iii) disseminate resulting understandings, information, knowledge, and experiences. The outcomes of this process could lead to cognitive products, such as changes in policy beliefs, whether in the direction of confirmation or revision, or behavioral products, such as policy changes or changes in policy advocacy strategies (Heikkila & Gerlak, 2013; Jones & Baumgartner, 2005b; Zaki et al., 2022). In the next section, we build on the model of information processing and policy learning developed by Nowlin (2021). We argue that the model explains the underlying mechanisms at play, at both the individual and collective level, for the learning process outlined above.

2.5 A Model of Policy Learning and Information Processing

Recent work combined policy-oriented learning with insights from the literature on information processing to develop a model of policy learning through information processing (Nowlin, 2021). The model is focused on cognitive processes associated with learning and uses a Bayesian framework to illustrate belief change or updating. Additionally, the model can be applied to learning by both individuals and collectives because the underlying processes exist at both levels. In brief, a Bayesian framework involves a prior belief, information that is processed through a likelihood function, and a posterior belief that is a function of the prior beliefs and the information. In more formal terms and using notation from Druckman & McGrath (2019), a prior belief is understood as a probability distribution about the true state of the world expressed as $\pi(\mu) \sim N(\hat{\mu}_0, \hat{\sigma}_0^2)$ where $\pi(\mu)$ is the true state of the world, $\hat{\mu}_0$ is an individual's belief about the state of the world, and $\hat{\sigma}_0$ is their uncertainty about their beliefs. Next, the individual encounters some information, x, about the true state of the world expressed as $N(\mu, \hat{\sigma}_x^2)$, where $\hat{\sigma}_x^2$ represents the individual's likelihood function or how she processes and weights the information. Finally, the individual updates her beliefs in light of the information $\pi(\mu|\text{x})$.

The steps of the learning through information processing model include:

- **Prior belief**: $\pi(\mu) \sim N(\hat{\mu}_0, \hat{\sigma}_0^2)$
- **Processing of information, x (likelihood function)**: $N(\mu, \hat{\sigma}_x^2)$
- **Posterior belief**: $\pi(\mu|\text{x})$

Under this model, learning only occurs when beliefs or belief certainty changes in the direction of the information (Nowlin, 2021, 2024). The model is about the process of learning and does not consider the validity of the beliefs or the information or how receivers of information process that information. As noted by Coppock (2023), “people are free to have bonkers prior beliefs and loopy likelihood functions” (123). Additionally, if individuals were perfectly rational and motivated solely for accuracy, then the posterior beliefs would be updated in direct proportion to the information. In other words, a “strong” information signal – one that is valid, credible, reliable, and reflects the true state of the world – would lead to shifts in beliefs that reflect the strength of the signal. However, individuals are not perfectly rational and, therefore, will weight the information, as noted by $\hat{\sigma}_x^2$, based on the strength of their prior beliefs (Nowlin, 2021). As a result, information and information processing may or may not lead to learning. Next, we discuss several possible learning outcomes that result from the learning process described above.

2.5.1 Learning Outcomes Related to Beliefs

The process of learning through information processing leads to several possible outcomes or products based on the motivation and biases of the individual or group. As noted, beliefs are probably distributions with a central tendency (the belief), $\hat{\mu}_0$, and uncertainty, $\hat{\sigma}_0^2$, about the belief. Changes in either the belief or certainty are a possible outcome; however, the changes must reflect the information to be considered learning.[2]

Possible learning outcomes associated with beliefs include:

- **No learning**: No change in beliefs or level of certainty
- **Belief decay**: A decrease in certainty

2 Another possible outcome is a back-lash or boomerang effect, where beliefs move in the opposite direction of the information (Druckman & McGrath, 2019; Hart & Nisbet, 2012). However, in our conceptualization, learning from information processing only occurs when the posterior beliefs are congruent with the information. Therefore, a boomerang effect would not be considered learning even though it is in response to information.

- **Belief reinforcement**: An increase in certainty
- **Belief change**: A change in the central tendency of the belief distribution

Additionally, belief changes may be coupled with decay and/or reinforcement as both the beliefs and the certainty change in response to information. For example, Nowlin (2024) finds that following a deliberative mini-public about several environmental issues, participants' beliefs changed in various ways on different topics. Participants experienced several possible outcomes, including no learning (no changes in beliefs or certainty), belief changes (a significant shift in the mean), and belief reinforcement (a reduction in uncertainty). Additionally, on some issues, participants saw belief change and reinforcement, where beliefs changed in conjunction with a decrease in uncertainty.

As noted, if individuals were comprehensively rational and motivated for accuracy, then they would update their beliefs in a way that is proportionate to the information. However, it is well known that individuals are not comprehensively rational and are often motivated to reach specific conclusions rather than accuracy, which makes belief change in response to information much less likely (Jones, 2017; Kunda, 1990). In addition, several possible biases reduce the likelihood that individuals or groups learn in light of information, including in-group/out-group, confirmation bias, desirability bias, availability bias, negativity bias (loss aversion), anchoring bias, and status-quo bias (Heikkila et al., 2024).

With the policy learning through information processing model, motivated reasoning and biased information processing are captured by the interaction between prior beliefs and the likelihood function. The likelihood function is how the information is processed and weighted, and information signals are processed and weighted disproportionately, with some signals ignored and others weighted strongly. In a Bayesian framework, the posterior belief is proportional to the prior belief multiplied by the likelihood function, meaning that a strong prior and/or a strong weighting of information signals will determine the degree to which beliefs change. This process likely underlies many common biases. For example, confirmation bias can be understood as the over-weighting of information that confirms prior beliefs and the under-weighting of information that challenges prior beliefs.

Additionally, the interaction of prior beliefs and the likelihood function varies by type of belief. Deep core beliefs would provide strong prior beliefs and act as filters to weight information. While not impossible, changing deep core beliefs would take an exceptionally strong information signal that was weighted appropriately. Policy core beliefs are also difficult to change and would also require a strong information signal. However, policy core beliefs may become eroded as different dimensions of an issue start to shift and become more (less) salient as a result of information. Learning regarding policy core beliefs can happen when an issue gets redefined. Finally, secondary beliefs and subsequent policy preferences are the beliefs most likely to change, as the prior beliefs associated with secondary aspects or policy preferences are the least strongly held.

2.6 Conclusion

Policy learning through information processing occurs among policy actors operating within and across institutions. In this chapter, we introduced policy-embedded learning as a conceptual approach that captures the importance of information and information processing in policy learning. Then, we discussed the nature of belief systems and policy-oriented learning. Next, we examined the literature on information processing and collective learning. Finally, we outlined a model of learning through information processing that underlies how individuals and/or collective learning (i.e., update beliefs) as a result of information and information processing. The discussion in this chapter sets the stage for the information processing and policy-embedded learning framework that we develop and examine in the next chapter.

3. The Information Processing and Policy-Embedded Learning Framework

3.1 Introduction

Policy actors and institutions generally have a wide array of information available about potential problems, policy alternatives, and political factors. All of this available information has the potential to contribute to policy learning. Yet, actors and institutions are not able to effectively process all of the available information because individuals do not have the capacity to effectively process all of the available and potentially relevant information. Therefore, learning is based on what bits of information receive attention and how that information is processed through the belief systems of policy actors and the institutional structures of the policymaking system. Policy learning through information processing occurs as information is processed and the beliefs of policy actors, individual or collective, are changed. The processing of information occurs as the various information signals are weighted relative to prior beliefs. Learning occurs when the posterior beliefs of policy actors shift and/or certainty shifts in the direction of the processed information. However, what is less clear are the possible mechanisms through which learning through information processing translates policy change.

In this chapter, we build on the approach from Chapter 2 to develop a framework of information processing, policy-embedded learning, and policy change. Within this framework, learning is embedded within a policy context where information is processed by policy actors within and across institutional systems and structures. The learning process is based on the learning through information processing model, and the process produces several possible learning products, including policy belief change, policy preference change, and behavior change, which leads to policy change. Finally, we empirically demonstrate the utility of this framework using public opinion data about nuclear energy in the United States of America.

3.2 Policy-Embedded Learning and Information Processing

The information processing and policy-embedded learning framework account for information, information processing by both policy actors and institutions, and learning outcomes that include cognitive (belief and preference change) and behavioral. Additionally, the framework includes pathways from policy learning (i.e., belief, preference, and behavior change) to policy change. Next, we discuss the various components of the framework, including information, information processing, learning, and outcomes.

3.2.1 Information

Information for policymaking is imperfect, costly, and asymmetric; however, policy actors face incentives to share information; otherwise, they risk allowing other actors to define the policy issue. Therefore, information about policy problems is ubiquitous in the policy process. It comes in the form of information about conditions and problems collected by government agencies (e.g., inflation, unemployment, crime), lobbying by interest groups, reporting by the media, and public opinion data, among many others. Additionally, focusing events, such as a disaster, terrorist attack, or pandemic, provides strong information signals about potential problems (Birkland, 1997, 1998, 2004; DeLeo et al., 2021). Information flows need to be processed to determine what, if any, actions need to be taken. However, given the amount of information present, only some information signals are processed, and many others are ignored.

3.2.2 Information Processing

The processing of information involves policymaking institutions, the systems and structures they create, and the policy actors involved in those institutions. Policy actors are limited in the amount of information that they are able to attend to and process, and their belief systems create biases regarding how the information to which they do attend is weighted. Institutions are structured to enhance information processing, for example, the ability of parallel processing of information; however, institutions can only take a few actions at a time. For example, legislative institutions can process information about multiple issues through a committee structure, yet they can only vote on one piece of legislation at a time. The systems and structures of institutional arraignments impose costs on information processing, such as decision and transaction costs. The information processing limits of individuals and institutions create friction that slows the path of information processing from learning to action, which leads to disproportionate information processing. The disproportionate processing of information leads to under- and over-reactions to information, a punctuated process of learning, and policy change, where problems are left unaddressed until a major policy change is needed.

3.2.3 Learning

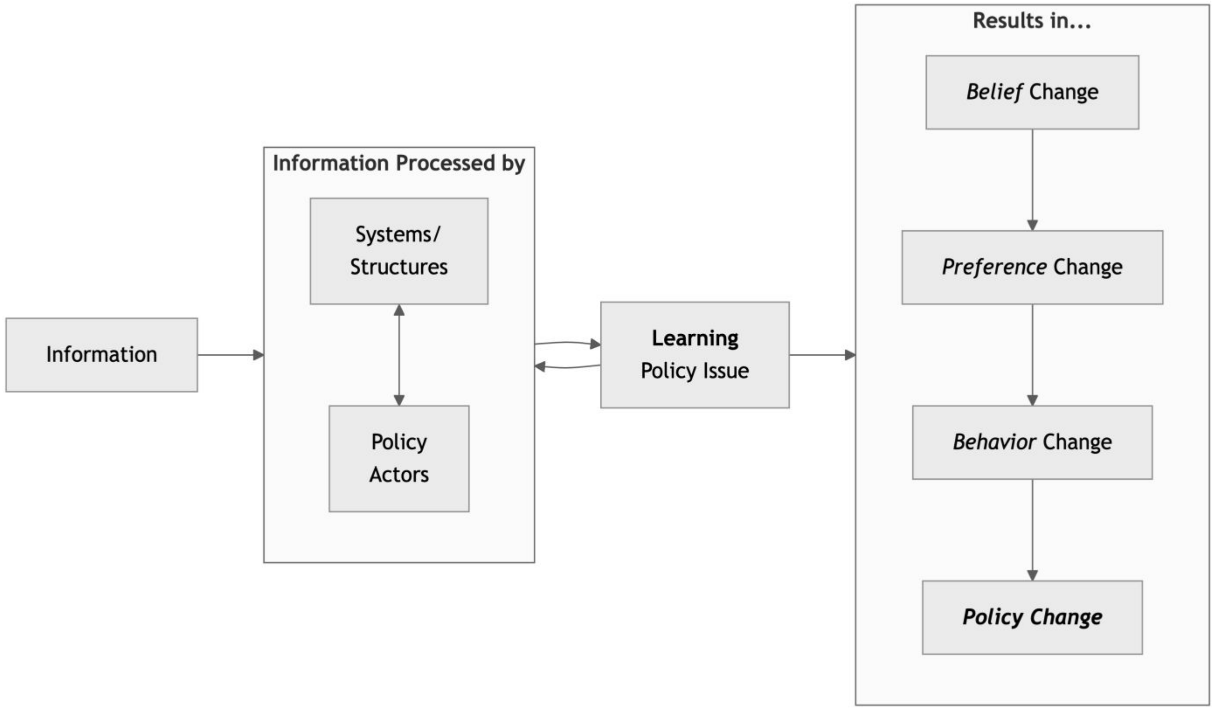
Policy learning through information processing occurs as beliefs are updated in response to information processed through cognitive and institutional structures. In policy-embedded learning, learning through information processing happens within a policy context and about a policy issue and focuses on policy core beliefs (Zaki, Wayenberg & George, 2022). Specifically, learning about policy issues involving policy core beliefs includes changes in how the issues are defined or understood. Policy issues contain multiple dimensions, and issue definitions involve how those dimensions are weighted, bundled, and placed together (Nowlin, 2016). Learning with regard to policy issues involves changes in the dimensions of the issue as a result of information and information processing. Like policy subsystems, issue definitions can reach static equilibrium points, and learning shifts that equilibrium as new dimensions are added or current dimensions are reconsidered.

3.2.4 Outcomes

As discussed in Chapter 2, several learning outcomes of the policy learning through information processing model are possible, including no learning, belief decay, belief reinforcement, and belief change. However, other outcomes are downstream or consequent to belief change, including policy preference change, behavior change, and, ultimately, policy change. Once belief change occurs, policy actors may update their policy preferences and engage in different behaviors or develop strategies to obtain their goals. Policy change may then arise as a result of those behaviors.

Putting these elements together, the information processing and policy-embedded learning framework is shown in Figure 3.1. As can be seen in Figure 3.1, information signals are processed by actors and institutions, which leads to learning about the policy issue. Learning can be incorporated by actors and institutions, such as when a policy issue is refined, and/or learning can lead to belief, preference, behavior, or, ultimately, policy change.

Figure 3.1: Information Processing and Policy-Embedded Learning Framework



The conditions under which policy learning leads to policy change have long been a central question, yet establishing how learning causes policy change has been elusive (Bennett & Howlett, 1992; Zaki, 2024b). Previous research has gained analytical traction on this issue by separating the process of learning from the products of learning, thus allowing insights into how learning processes connect to learning products (or outcomes) such as policy change (Heikkila & Gerlak, 2013; Zaki et al., 2022). More recently, Plümer (2024) developed an approach that examines three elements of learning products, including "(1) whether the observed learning product reflects a change in policy beliefs, (2) whether it signifies a change in policy preferences, and (3) whether it results in a change in policy outputs." We build on this approach, but we also include behavior change as a central feature of learning products that makes policy change more likely.

The learning products of belief, preference, and behavior change create distinct pathways to policy change. The product of belief change involves information processing and change in the direction of information, and preference change involves changes in the goals policy actors hope to obtain (Moyson, 2017; Plümer, 2024). Then, policy actors may change or adapt new behaviors as policy beliefs and preferences change. Additionally, behavior change may involve "new or enhanced informal routines and strategies, … new or expanded programs and plans that structure group behavior, or highly formalized rules or sets of institutional arrangements and policies" (Heikkila & Gerlak, 2013, pp. 491–492).

The causal pathways from policy learning to policy change include three possible outcomes: no policy change, non-congruent policy change, and congruent policy change (Plümer, 2024). No policy change results when only beliefs change or only beliefs and preferences change without a subsequent policy change. Non-congruent policy change happens when policy change occurs but is not aligned with the changed beliefs and preferences. The final pathway noted by Plümer (2024) is congruent policy change, where policy change matches the changed policy beliefs and preferences. To this approach, we add congruent and non-congruent behavior change that precedes policy changes. Non-congruent behavior change involves changes in behaviors that do not align with updated beliefs and preferences, whereas congruent behavior change does align with what was learned.

Table 3.1: Pathways of Policy Learning and Policy Change

Belief Change	**Preference Change**	**Behavior Change**	**Policy Change**
X			
X	X		
X	X	X (non-congruent)	
X	X	X (non-congruent)	X (non-congruent)
X	X	X (congruent)	
X	X	X (congruent)	X (congruent)

Adapted from Plümer (2024), Table 3.1 shows the various pathways through which policy change may happen as a result of policy learning through information processing. As each learning product occurs, the likelihood of policy change through policy learning increases; however, non-congruent changes in behavior and policy are also likely as changes may not align with updated beliefs and preferences.

In the next section, to illustrate the utility of this framework, we empirically explore the case of public opinion about nuclear energy before and after the Fukushima nuclear accident. The accident in Fukushima, Japan, was a focusing event that acted as a strong information signal about the risks of nuclear energy. In this case, we examine how policy core beliefs, specifically the issue definition of nuclear energy, and policy preferences changed as a result of Fukushima.

3.3 Application of the Information Processing and Policy-Embedded Learning Framework

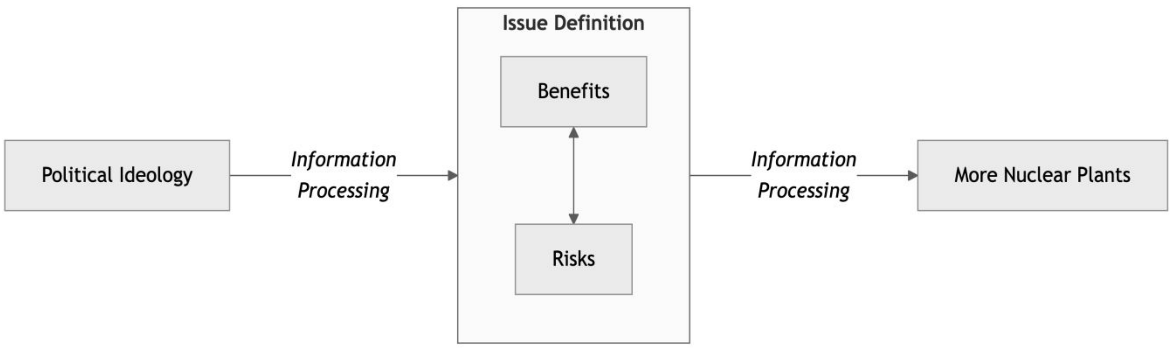
3.3.1 The US Public and Nuclear Energy

In March 2011, a large tsunami hit the coast of Japan. The nuclear plant power in Fukushima was flooded, which led to a loss of power in the plant and a subsequent release of radioactive material. Like the two previous nuclear accidents, Three Mile Island and Chernobyl, the accident at the nuclear power plant at Fukushima provided a potentially strong information signal about the risks of nuclear energy, and public support for nuclear energy tends to fall in the wake of such accidents (Besley & Oh, 2014; Gupta et al., 2019). Focusing events, such as accidents at nuclear power plants, can make it more likely that those issues reach the policymaking agenda (Birkland, 1998); however, focusing events and the information signals they send also creates opportunities for policy learning.

Using the information processing and policy-embedded framework discussed above, the Fukushima accident is a strong information signal about nuclear energy, and policy actors process that signal. Yet, the signal is filtered through their belief systems, and as laid out in the model of learning through information processing, learning occurs when beliefs are updated in the direction of the information. As noted, policy actors are assumed to have a three-tiered belief system that includes deep core beliefs, policy core beliefs, and secondary beliefs. Within the belief hierarchy, beliefs move from abstract foundational beliefs – deep core beliefs – to beliefs specific to a policy issue or subsystem – policy core beliefs – to secondary beliefs, which are about specific means to achieve policy goals. Deep core beliefs inform and constrain policy core beliefs, and both deep core and policy core shape secondary beliefs. The model of learning through information processing focuses on the updating of beliefs resulting from the processing of policy-relevant information. While learning as belief updating could occur for any type of belief, as beliefs become increasingly applicable to multiple policy issues, they become less likely to change. It is rare that deep core beliefs such as political ideology or cultural worldviews change, whereas narrowly focused secondary beliefs are the most likely to change. Policy core beliefs are more likely to change than deep core beliefs but less likely than secondary aspects. One way for learning to occur regarding policy core beliefs is through a shift in the issue definition that undergirds policy core beliefs.

At the heart of policy core beliefs is the issue definition, or the various dimensions of the issue weighted by their salience. As dimensions are added or the salience of the dimensions changes, learning regarding policy core beliefs becomes more likely. This is illustrated in Figure 3.1, as learning about a policy issue involves changes to the issue definition. With the model of policy learning through information processing, learning occurs when the information signal is strong relative to the prior beliefs. Strongly held prior beliefs would only change with an extremely strong and heavily weighted information signal. In the case of nuclear energy, the Fukushima accident was a strong signal about the risks of nuclear energy and likely made the dimensions of risks associated with nuclear energy more salient, thus altering the issue definition and thereby alternating the policy core beliefs. As shown in Table 3.1, policy preferences may change following the updating of policy core beliefs. This process for nuclear energy is shown in Figure 3.2.

Figure 3.2: Ideology, Nuclear Energy Beliefs, and Preferences for More Nuclear Plants



As shown in Figure 3.2, information is processed relative to deep core beliefs (i.e., political ideology), and as a result, the issue definition of nuclear energy (i.e., policy core beliefs) is potentially updated. Note that the issue definition of nuclear energy is constituted by dimensions associated with the benefits and risks of nuclear energy. The updating of policy preferences results from the same learning dynamic. Information processing occurs along the chain of deep core beliefs → policy core beliefs → policy preferences, and updating occurs when the information signal is sufficiently strong relative to the prior belief or preference. Learning happens if the beliefs or the certainty associated with the beliefs change significantly. To estimate the model shown in Figure 3.2 we use survey data of the US public collected over 17 years.

Data and Measures

To examine the impact of the Fukushima accident on beliefs about nuclear energy, we draw on data from 17 years, 2006-2023, of nationwide surveys of US residents.[3] In each survey, respondents were asked about their views on energy and environmental issues, including nuclear energy. Additionally, the surveys gauged the deep core beliefs of respondents by including measures of political ideology. The Institute for Public Policy Research and Analysis at the University of Oklahoma developed and administered the surveys and fielded them annually. The samples for each survey were recruited by firms including Survey Sampling International and Qualtrics, and quotas were used to ensure that the samples were representative of the US population. However, our interest is in policy learning among elite policy actors, not the general public. To best account for how policy elites might update beliefs and preferences, we use only the sample of survey respondents who have completed college. Overall, the combined surveys include 15311 respondents. Next, we discuss the various measures we used to examine the dimensions of nuclear energy and policy preferences associated with nuclear energy.

Deep Core Beliefs

For deep core beliefs, we use a measure of self-reported political ideology. Among the US public, support for nuclear energy is polarized along ideological lines, with conservatives, on average, more supportive of nuclear energy than liberals (Besley & Oh, 2014; Hawes & Nowlin, 2022; Kuklinski et al., 1982; McBeth et al., 2023). To measure political ideology, respondents were asked, "On a scale of political ideology, individuals can be arranged from strongly liberal to strongly conservative. Which of the following categories best describes your views?" with 1

3 Some years are missing due to question wording changes.

indicating strongly liberal, 4 = middle of the road, and 7 = strongly conservative. The mean on the political ideology scale is 3.969, which is the middle of the road.

Policy Core Beliefs: The Dimensions of Nuclear Energy

To explore the issue dimensions of nuclear energy, we rely on a set of questions regarding the risks and benefits of nuclear energy. The assumption is that views about the risks and benefits are the key drivers of how the issue of nuclear energy is defined and that the weighting of those dimensions shapes the policy core beliefs of the nuclear energy issue. Respondents were asked to rate, on a zero to ten scale, their perceived risks of nuclear energy across four dimensions of risk, including an accident at a nuclear power plant, an accident during transportation or storage of nuclear materials, a terrorist attack on a nuclear plant, and diversion of fuel to make a nuclear weapon. The exact question wording was as follows:

> First we want to know about your beliefs concerning some of the possible risks associated with nuclear energy use in the U.S. Please consider both the likelihood of a nuclear event occurring and its potential consequences when evaluating the risk posed by each of the following on a scale from zero to ten, where zero means no risk and ten means extreme risk. [*Statements were presented in random order*]

- **Plant Accident**: An event at a U.S. nuclear power plant within the next 20 years that results in the release of large amounts of radiation
- **Transportation Accident**: An event during the transportation and temporary storage of spent nuclear fuel from nuclear power plants in the U.S. within the next 20 years that results in the release of large amounts of radiation

- **Terrorist Attack**: A terrorist attack at a U.S. nuclear power plant within the next 20 years that results in the release of large amounts of radiation
- **Weapon**: The diversion of nuclear fuel from a nuclear power plant in the U.S. within the next 20 years for the purpose of building a nuclear weapon

Similarly, respondents were asked about the benefits of nuclear energy across four dimensions, including no greenhouse gases, a source of reliable power, contributions to US energy independence, and reduced environmental damage compared to coal or oil and gas extraction.

Next we want to know about your beliefs concerning some of the possible benefits associated with nuclear energy use in the U.S. Please evaluate each of these possible benefits of nuclear energy use on a scale from zero to ten, where zero means not at all beneficial and ten means extremely beneficial. [*Statements were presented in random order*]

- **No GHG**: Reducing environmental threats because the generation of nuclear energy does not produce greenhouse gases that are believed to cause climate change
- **Reliable Power**: Reliable power because nuclear energy generates large amounts of electricity and is not affected by weather conditions, such as low rainfall or no wind
- **Energy Independence**: Greater U.S. energy independence because nuclear energy production does not require oil or gas from foreign sources
- **Reduced Environmental Damage**: Reduced environmental damage because of less need for mining coal or extracting oil and gas

Policy Preferences

To measure policy preferences, we examine support for building more nuclear power plants in the US. Specifically, we averaged responses to two questions, with one regarding support for adding additional nuclear reactors at existing nuclear power plants and the second for adding new reactors at new locations in the US.[4]

> Using a scale from one to seven, where one means *strongly oppose* and seven means *strongly support*, how do you feel about constructing additional nuclear reactors at the sites of existing nuclear power plants in the U.S.?
>
> Using the same scale from one to seven, where one means *strongly oppose* and seven means *strongly support*, how do you feel about constructing additional nuclear power plants at new locations in the U.S.?

For analysis, we combine and average the two scales into a single one to seven measure of support for more nuclear energy plants. The average support for more nuclear plants is 4.28.

Results

We first examine differences in means of the dimensions of nuclear energy – the measure of risks and benefits – pre-and-post the Fukushima accident. The pre-Fukushima measures include pooled survey data from 2006 to 2011, and post-Fukushima is pooled survey data from 2012 to 2023. Additionally, we combined and averaged the risk and benefits items into single scales. The Cronbach's α for the risk items was 0.922, and the α for the benefit items was 0.906. Table 3.2 shows the pre-and-post-Fukushima means for each dimension and the averaged items.

4 We do not have a measure of secondary beliefs, rather we assume that secondary beliefs are implied by policy preferences.

Table 3.2: Comparison of Nuclear Energy Issue Dimensions Pre-and-Post-Fukushima

Scale	Issue Dimensions	Pre-Fukushima Mean	Post-Fukushima Mean
Risks (0 to 10 scale)	Plant Accident	5.78	6.37***
	Transportation Accident	5.85	6.28***
	Terrorist Attack	6.37	6.58***
	Weapon	5.26	5.77***
	All Items	5.82	6.26***
Benefits (0 to 10 scale)	No GHG	7.25	7.02***
	Reliable Power	7.37	7.25**
	Energy Independence	7.513	7.3***
	Reduced Environmental Damage	7.21	7.17
	All Items	7.35	7.195***

As shown in Table 3.2, other than reduced environmental damage, all of the dimensions were significantly different post-Fukushima, as well as the averages of the risk and benefit items. The mean of each risk measure increased, and the mean of each benefit measure decreased following Fukushima. Also of note is that the average of the benefit measures had a significantly higher mean than the average of the risk measures both pre-and-post Fukushima. Pre-Fukushima, the average of the risk measures was 5.82, and the average of the benefit measures was 7.349, and the difference was statistically significant [t=-31.18, p<0.001]. Following Fukushima, the mean of the risk measures was 6.264, the mean of the benefit measures was 7.195, and the difference was significant [t=-27.25, p<0.001].

Next, we conduct an OLS regression analysis to examine the influence of the political ideology and the Fukushima accident on the issue definition of nuclear energy, specifically the perceived risks and benefits. Then, we explore the impacts of ideology, Fukushima, and the risks and benefits of nuclear energy on support for more nuclear power. Note that we also control for demographics, including age, gender (white=1), and race (white=1). The results are shown in Table 3.3.

Table 3.3: OLS Analysis of Political Ideology, Fukushima, Nuclear Energy Risks and Benefits, and Support for More Plants

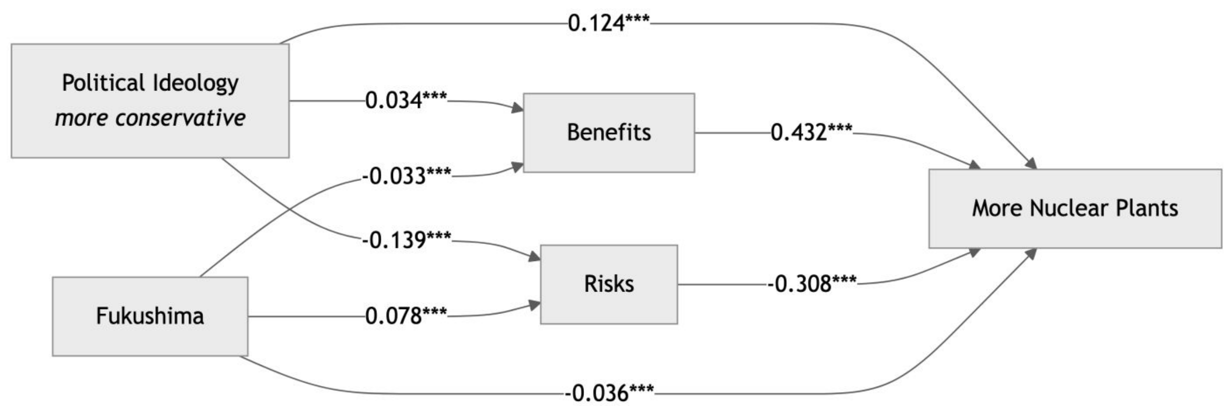
Dimension	Risks	Benefits	More Plants
Ideology (more conservative)	-0.156***	0.041***	0.124***
	(0.012)	(0.010)	(0.007)
Fukushima	0.447***	-0.153***	-0.136***
	(0.046)	(0.039)	(0.025)
Benefits	0.025*		0.355***
	(0.010)		(0.006)
Risks		0.017*	-0.202***
		(0.007)	(0.005)
Age	-0.026***	-0.001	-0.004***
	(0.001)	(0.001)	(0.001)
Gender (male=1)	-0.929***	0.483***	0.570***
	(0.043)	(0.036)	(0.024)
Race (white=1)	-0.482***	0.242***	0.084**
	(0.054)	(0.045)	(0.030)
N	14097	14097	13524
R^2	0.10	0.02	0.39

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

As can be seen in Table 3.3, increasing conservatism is associated with decreased risk perceptions and increased perceptions of benefits as well as support for more nuclear plants. In addition, perceptions of risks increased and perceptions of benefits and support for more plants both decreased following the Fukushima accident. Interestingly, the benefits and risks of nuclear energy are positively associated. But as expected an increase in the perceived risks is negatively associated with support for more plants, whereas an increase in perceived benefits is positively associated. For the demographic variables, age is negatively associated with both perceived risks and support for more plants. Finally, both males and whites perceive less risk, more benefits, and higher levels of support for more nuclear power plants compared to non-males and non-whites.

Our final analysis is a path analysis to show the influence of political ideology, Fukushima, and beliefs about the risks and benefits of nuclear energy on preferences for more nuclear power plants.[5] Path analysis includes standardized coefficients, which allow us to compare the impact of the various independent variables on the dependent variable. In addition, path analysis accounts for the various effects of political ideology and Fukushima on preferences for more nuclear power plants. Figure 3.3 shows the results of the path analysis, including the standardized betas.

Figure 3.3: Path Analysis of Ideology, Fukushima, Nuclear Energy Beliefs, and Preferences for More Nuclear Plants



As shown in Figure 3.3, increasing conservatism is associated with an increase in the perceived benefits of nuclear energy, a decrease in the perceived risks associated with nuclear energy, and an increase in the support for more nuclear power plants. Additionally, the Fukushima accident is associated with a decrease in perceived benefits, an increase in perceived risks, and a decrease in support for more nuclear plants. However, comparing the standardized coefficients, we see that ideology, and Fukushima had roughly the same influence on perceived benefits (0.034

5 For the path analysis, we used the standardized coefficients from the OLS regression, as shown in Table 3.3.

vs. -0.033); however, ideology had a larger influence on perceived risk than Fukushima (-0.139 vs. 0.078) and a larger influence on support for more plants (0.124 vs. -0.036).

Path analysis also allows us to examine the effect of political ideology on support for more nuclear plants (*direct effects*), the mediation effect of political ideology operating through the issue definition (*indirect effects*), and the combined effects of political ideology and the issue definition of nuclear energy on support for more nuclear plants (*total effects*). The indirect effect is calculated by multiplying the path (a) from political ideology to the issue definition and the path (b) from the issue definition to support for more plants. The effects are shown in Table 3.4.

Table 3.4: The Effects of Political Ideology, Fukushima, and the Risks and Benefits of Nuclear Energy on Support for More Nuclear Plants

	Indirect Effect	Direct Effect	Total Effect
Increasing Conservatism and Nuclear Risks	0.043	0.124	0.167
Increasing Conservatism and Nuclear Benefits	0.015	0.124	0.139
Fukushima and Nuclear Risks	-0.024	-0.036	-0.06
Fukushima and Nuclear Benefits	-0.014	-0.036	-0.05

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

As shown in Table 3.4, deep core beliefs had larger total effects than the Fukushima accident on support for more nuclear power plants. These results illustrate the influence of deep core beliefs on policy preference even when learning occurs regarding policy core beliefs.

4. Concluding remarks: What this element does, and what comes next

Revisiting conceptual, theoretical, and analytical approaches to policy learning

In the first chapter of this element, we started with the question of why, how, and when policy actors do what they do, a question the answer to which has long been a central quest for public policy scholarship. Among the different theoretical approaches that were developed to answer this question, policy learning stands out as one of the most omnipresent and potent. In reviewing learning literature, we illustrated how, in an attempt to answer the aforementioned question, policy learning research has grown to become an active research field sprawling with various conceptual and theoretical approaches underpinned by different ontological-epistemological positions from mechanistic, positivist, interpretivist, to social constructivist (Zaki, 2024b; Zaki & Radaelli, 2024). With such rich and diverse foundations, various theoretical descriptors of policy learning processes have emerged, resulting in a wide range of types, mechanisms, forms, modes, and degrees of learning. Those were developed through – and deployed in a wide range of analytical approaches scaling up and down between micro-foundations and individual learning dynamics, collectives, and systems (see Heikkila & Gerlak, 2013; Dunlop & Radaelli, 2017; Nowlin, 2021; Zaki & Wayenberg, 2023). There, and as shown in Table 1.1, we see learning being analytically employed in various ways, each anchored in specific conceptual approaches, from mechanistic and process-oriented to paradigmatic, ideational, and outcome-oriented.

This theoretical versatility of policy learning has led to it being viewed as an ontology of the policy process, which is being recognized as providing a coherent and robust explanatory view of why policy actors do what they do, as a function of how, why, and when they learn about policy problems (see Kamkhaji 2022). This is premised on an overarching understanding of the learning process as "policy-embedded," where policy actors seek and/or process information and knowledge about policy problems while being embedded within a particular policy context, and where the information and knowledge they collate interacts with different institutional structures therein (see Zaki, Wayenberg & George, 2022). This policy-embedded learning process is accordingly viewed as the driver for different policy behaviors, regardless of normative assessments of the quality of learning outcomes. For example, being traceable to updates in policy-issue understandings, beliefs, and potentially, from there on, updates in policy-related behaviors, from advocacy to varying degrees of policy change. By adopting this micro-foundational perspective, policy learning is seen as the kernel and nucleus of individual and collective behavior. We then carried this fundamental approach to policy learning over to chapter two, pursuing the second contribution of this element, to which we now turn.

A deeper understanding of policy learning dynamics: An Information Processing approach

In Chapter Two, we pivoted from the background conceptualization of policy-embedded learning to examine the micro-foundations of how learning occurs. This is by tracing and analyzing the role of information – the building block or raw material of the learning process – and how it interacts with pre-existing belief structures. To do so, we constructed a model of policy-embedded learning and information processing. This model focuses on belief change as the main learning outcome, operationalizing four varieties as a function of the interaction between pre-existing beliefs and inbound information, including no learning, belief decay, belief reinforcement, and belief change. In addition to belief, the model suggests that preference change, behavior change, and policy change are outcomes of policy learning.

After developing the policy-embedded learning and information processing model, in Chapter Three, we placed the micro-foundational model within the broader context of policymaking, which includes information, information processing by policy actors and institutions, and policy learning. We also discussed pathways from policy learning to policy change, which includes belief change, preference change, behavior change, and policy change. For policy change to happen as a result of policy learning through information processing, all elements must be consistent and consistent with the information. This means that belief change must be consistent with the information, preference change must be consistent with belief change, behavior change must be consistent with belief and preference change, and policy must be consistent with belief, preference, and behavior change. We then used many years' worth of public opinion data from the US about nuclear energy. We leveraged this data to explore the process of learning regarding updating beliefs and preferences following the information signal of the risks of nuclear energy provided by the 2011 accident at the nuclear power plant in Fukushima, Japan. That example showed belief change regarding the risks and benefits of nuclear energy as well as preference change regarding the construction of new nuclear power plants congruent with the information.

What comes next? Avenues for future research

Having discussed our model of policy-embedded learning and information processing, we now turn to avenues for future research. The first avenue concerns the analytical level scalability of this model. As earlier noted, learning occurs within and across various levels, micro, meso, and macro, spanning individuals, collectives (such as advocacy coalitions, policy subsystems, interest groups, and organizations) as well as systemically. Furthermore, learning not only occurs within these levels but also travels across them (Heikkila & Gerlak, 2013; Dunlop & Radaelli, 2017; Zaki, Wayenberg & George, 2022). Banking on the interactions between clear structural dimensions of policy actors, information and knowledge, and structures (whether cognitive or institutional), the policy-embedded learning model is highly scalable across levels. As such, future research can apply this model across multiple levels of analysis, for example, within particular collectives such as networks, advocacy coalitions, or organizations where the underlying mechanisms of interaction between prior beliefs, information processing, and weighted posterior beliefs are assumed at each level. Furthermore, and in terms of analytical level-scalability, this approach could also be applied across levels as, for example, tracing how individual learning connects to coalition learning, which then connects to subsystem learning.

The second avenue for future research concerns theoretically scaling up this model and embedding it within other theoretical perspectives of learning to enhance their explanatory potential. This could include various types, modes, mechanisms, and forms of learning. For example, this includes empirically tracing instrumental learning processes (mostly focused on policy instrument calibrations) through analyzing the processing of information related to secondary beliefs. Similarly, future research can pursue the same approach within other modes of learning, whether epistemic, hierarchical, reflexive, or bargaining-oriented. This also extends to learning mechanisms, whereby the interaction between information and cognitive structures can be traced within inferential learning, where information signals are carefully considered and weighted, versus the case of contingent learning, where crises or focusing events quickly shift how issues are defined and reshuffle policy priorities.

Corollaries and limitations

While the focus of this Element has been policy learning, there are several other ways in which policy change may occur. It is important to note the limits of our approach as an explanation for policy change. ACF scholarship notes that apart from policy-oriented learning policy change, both major (change associated with policy core) and minor (change associated with secondary aspects) can result from events both external and internal to the policy subsystem (Nohrstedt et al., 2023). Events or shocks can include changes in socioeconomic conditions, shifts in public opinion, changes in governing arrangements, elections, scandals, policy failures, or crises or disasters. Such events can constitute a strong information signal that leads to policy learning, such as in the cases of the Fukushima accident, the COVID-19 pandemic, or the Russian invasion of Ukraine, among others. However, apart from learning, events can lead to policy change through shifts in the resources available to coalitions or when a coalition pushing for policy change is able to exploit a policy window that opens as a result of the event (Nohrstedt & Weible, 2010).

The ACF also notes that policy change can happen as a result of a negotiated agreement between coalitions (Sabatier & Weible, 2007). Negotiated agreements involve deliberation, compromises, and trade-offs between adversarial coalitions, which result in policy changes. Such agreements are more likely in the presence of a "hurting stalemate" or a situation where the status quo is unacceptable to the coalitions, and there are no other policy venues in which to engage (Nohrstedt et al., 2023). Finally, policy change can happen as a result of shifts in power between coalitions (Pierce et al., 2020). Power shifts between coalitions can happen as a result of events such as an election that favors elected officials of one coalition, newly available resources for a coalition such as a powerful elected official, or knowledgeable experts (Nowlin et al., 2022), among others.

While information processing is at the heart of our approach to learning in this element, policy change can also happen as a result of attention to information without any changes in beliefs or preferences (Workman et al., 2009). Policy actors have multiple and perhaps competing policy preferences. Policy change can happen when one preference becomes more salient than another in the absence of learning or updating preference. One example from Jones (1994) is the consideration of the US Congress in 1991 and again in 1992 to fund the construction of the superconducting super collider, a partial accelerator that would have been built in Texas. Seventy-nine US House of Representatives members voted in favor of funding the super collider in 1991 but subsequently voted against funding in 1992. The shift in votes by the Seventy-nine members of Congress was likely a result of the issue of the budget deficit gaining more attention and making the preference for a smaller deficit (and perhaps re-election) more salient (Jones, 1994).

In this element, we started with a review and mapping of the different theoretical directions in policy learning literature, illustrating the potential for integrating an information processing approach therein. This has demonstrated the substantial promise of the information processing approach to enhancing the analytical utility of policy learning research. By tracing how information is processed and reconciled against cognitive and institutional structures, this approach offers a more nuanced understanding of the micro-foundations of policy learning. It also helps advance the analytical utility of different theoretical approaches to learning. By acknowledging the cognitive processes that underpin policy decisions, scholars can gain deeper insights into the dynamics of policy change and the factors that influence learning across different contexts.

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